CS1200 / CSCI E-120 Lecture 15

Embedded EthiCS

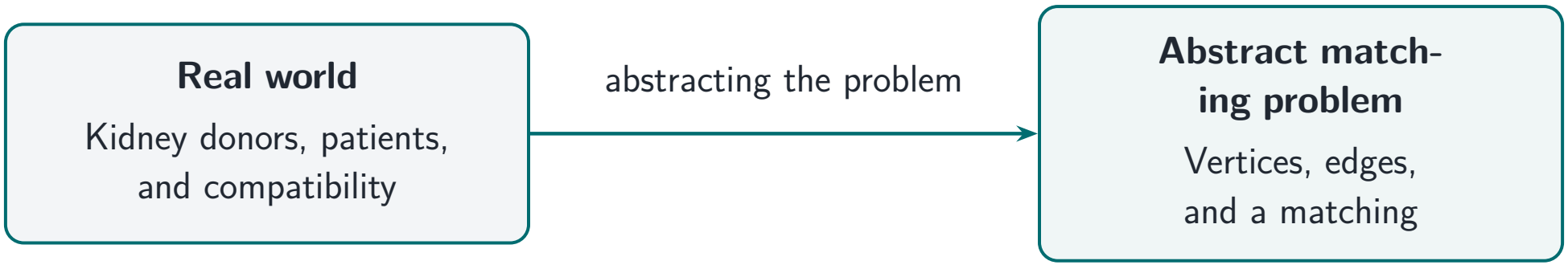
Ethical and Social Considerations

Fall 2026

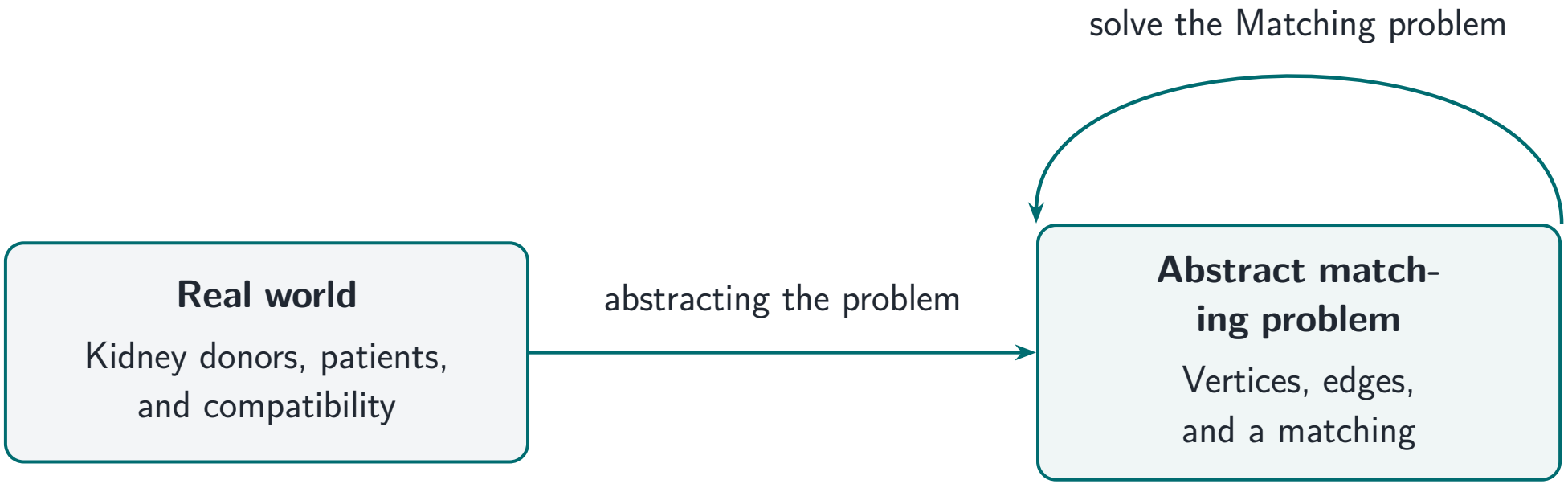
Announcements

- OpenAI+Anthropic: Exponential Time Hypothesis is false via faster than expected algorithm for 3SAT!! We ll do 3SAT in 2 weeks and briefly discuss this breakthrough.
- Pset 4 in-class reflection: very few pset scores were dropped; so most of the in-class reflections "reflect" the actual understanding at submission (which is great!).
- Midterm 2 next Monday! Bring black or blue pen (avoid pencil).
- Pset 6 released - due well after midterm (Nov 12).
- No pset 5 in-class reflection.
- Anurag OH after class.
- Guest today: Matt Kopec (Program Director for Harvard's Embedded EthiCS program.)

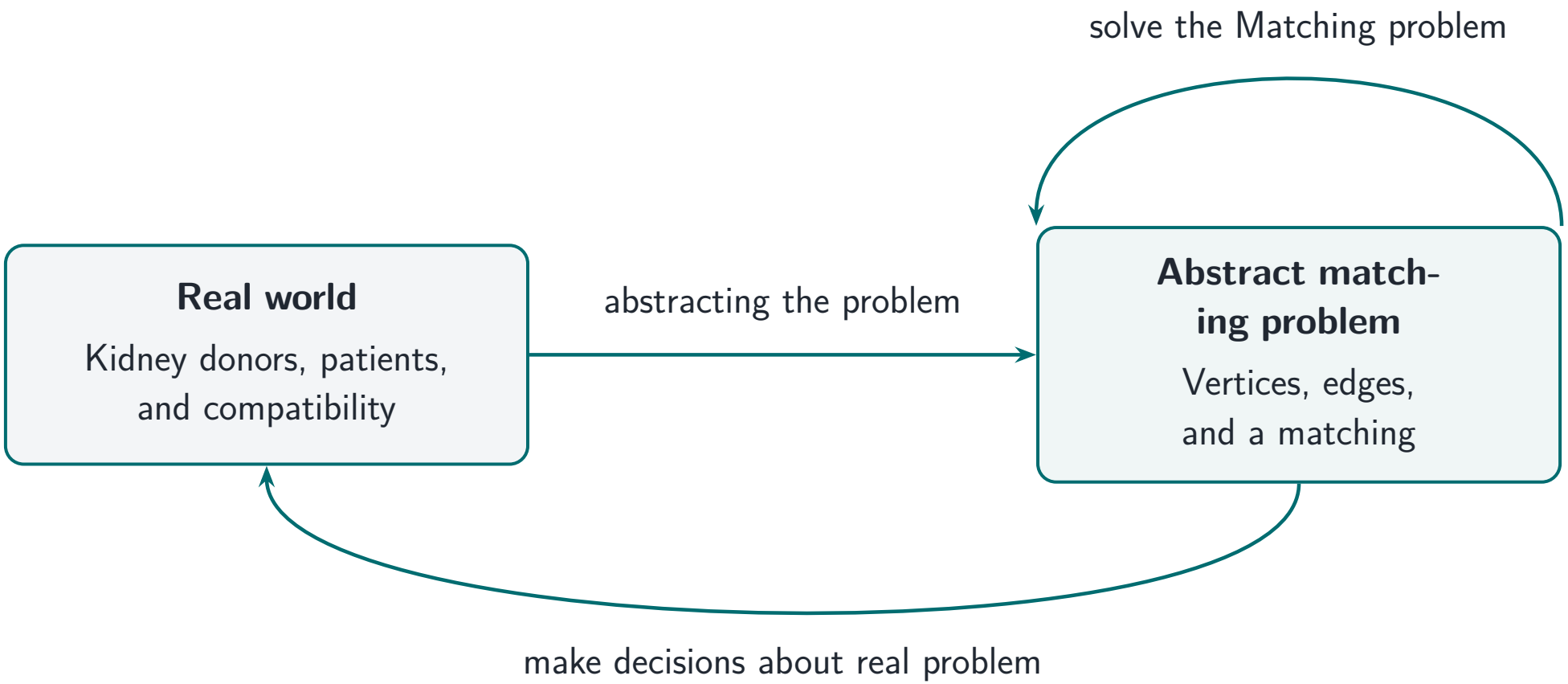
Interrogating our Mathematical Models



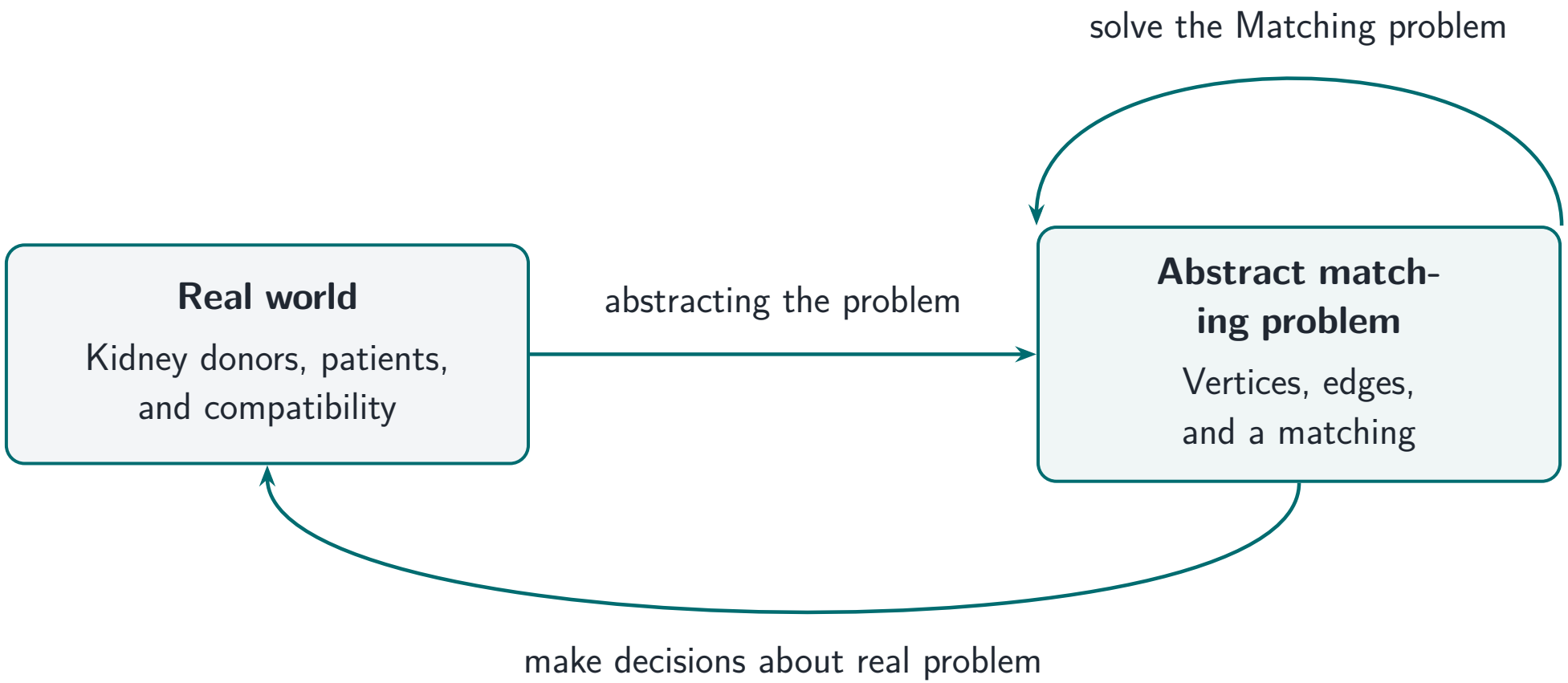
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Q: what factors are missed in this abstraction?

Interrogating our Mathematical Models

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- **Urgency:** some patients may not survive a long wait.
- **Prior organ donation:** we may want to give extra priority to someone who previously donated an organ.
- ...

Concrete issues

- **Scarcity:** few donors and many potential recipients (90,000 on waitlist in Sep 2024 and 30,000 transplants done in 2023). Thus, there may be too many solutions to the Matching problem – how do we decide?

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- **Priority:** Scarcity means we need to give priority to some patients. What is our criteria?
- **Maximizing 'good':** We would like to maximize 'amount of good', by making a choice of who receives a kidney.

Specifying the Problem

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Patient feature	**Numerical representation**
Probability of survival	
Previous living organ donation	
Expected life-years	

Q: What could be represented as a bit, a rational in $[0, 1]$, a nonnegative rational?

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Maximum Vertex-Weighted Matching

Let $V(M)$ be the set of vertices matched by M.

$$w(M) = \sum_{v \in V(M)} w(v) = \sum_{\{u,v\} \in M} \big(w(u) + w(v)\big).$$

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Choice of those weights and the summation itself require ethical reflection.

Is sum over weights the 'right choice'?

Consider two patients who are identical in all ethically relevant respects. Suppose our ethical judgment ranks the outcomes as follows:

$$\text{both matched} \succ \text{neither matched} \succ \text{exactly one matched.}$$

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More broadly, choosing to focus on *quantifiable* values is itself a substantive choice. Some ethical approaches require judgments that a numerical objective may not capture.

Utilitarianism

A guiding idea: choose the action expected to produce the best outcome or welfare.

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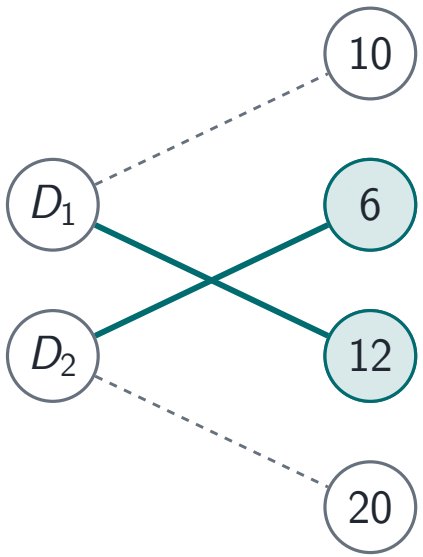
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Equal Value of Life Years

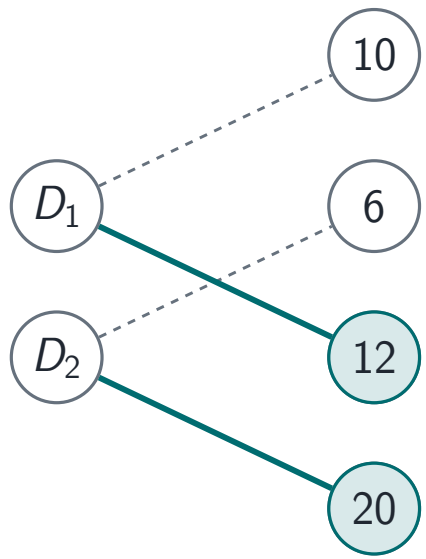
eVLYs: count the expected life-year gained - assuming each life-year gained is same for all individuals.

Patient labels (right) below are life-years gained; donor (left) weights are 0.

Matching M



Matching M'

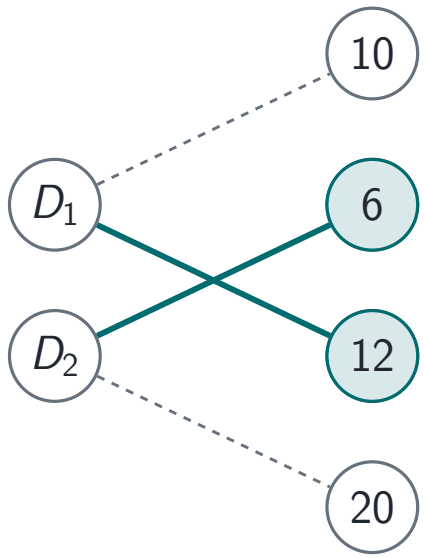


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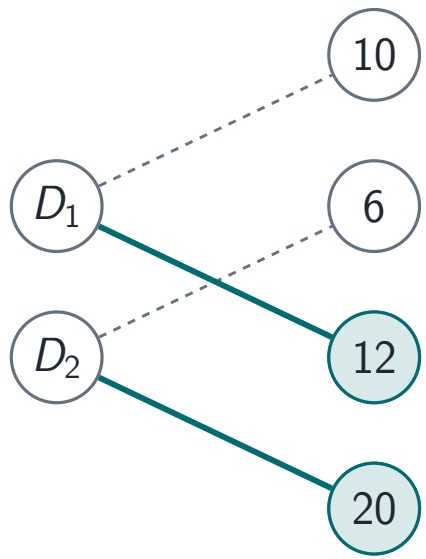
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Matching M



$w(M) = 6 + 12 = 18.$

Matching M'



$w(M') = 12 + 20 = 32.$

Equal Value of Life Years: Concerns

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eVLY still prefer M': the metric treats all gained years equally.

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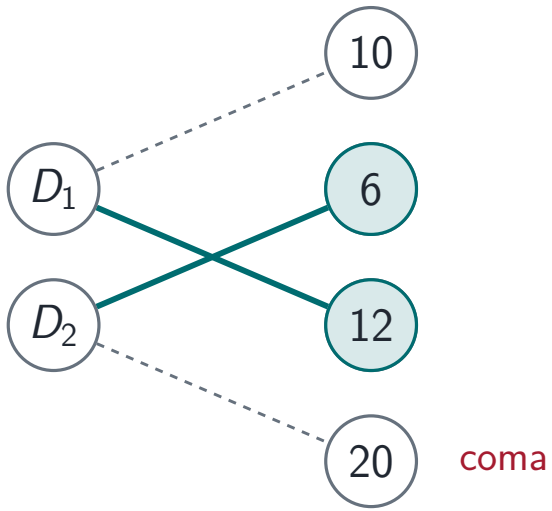
How is q determined? One way is to survey public's preferences across years in different health states.

Q: how does QALY address the problematic eVLY example?

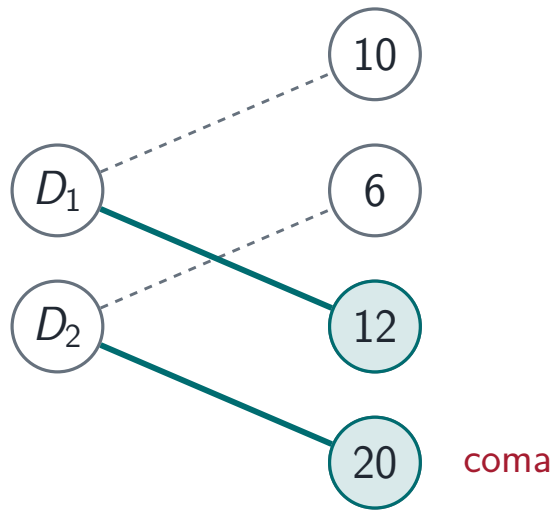
Quality-Adjusted Life Years: The Example

Near-perfect health has weight approximately 1, and a year in a coma has weight approximately 0.

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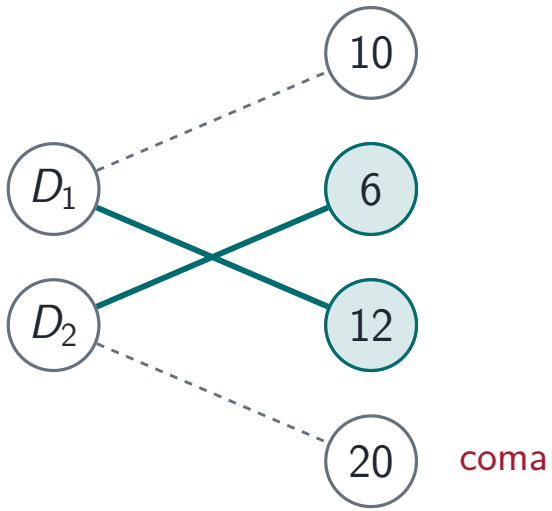
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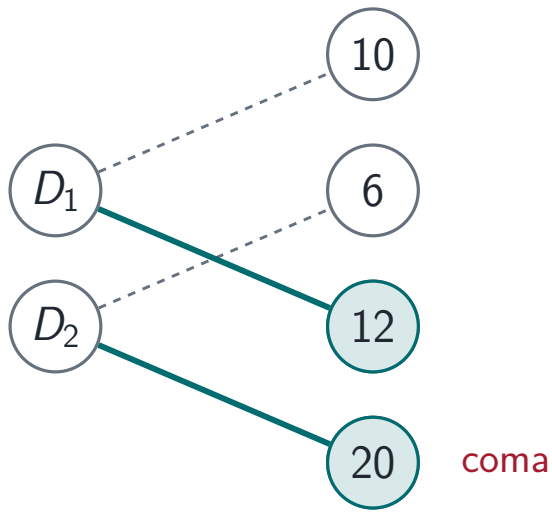
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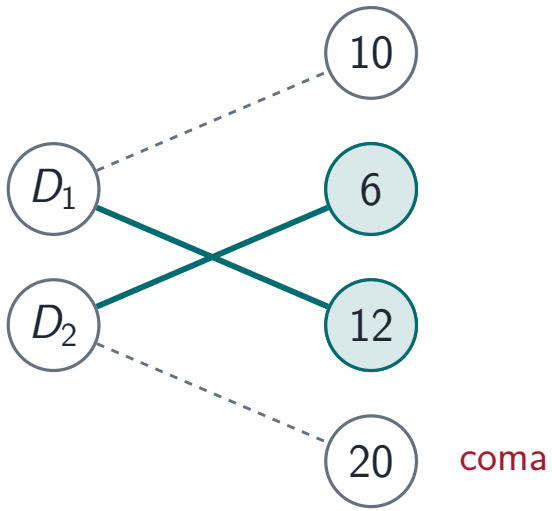


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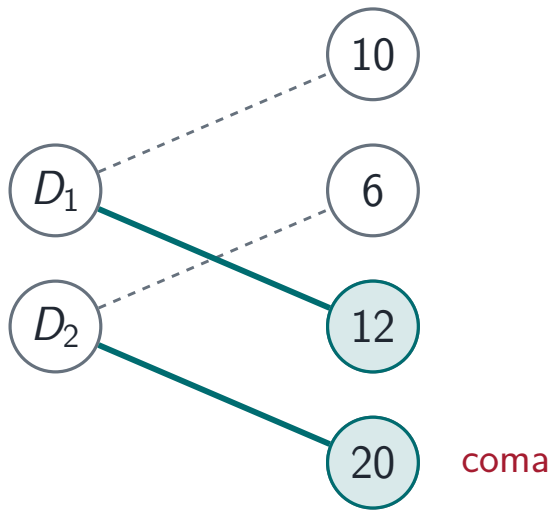
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Under these assumptions, maximizing QALYs leads to choosing M over M'.

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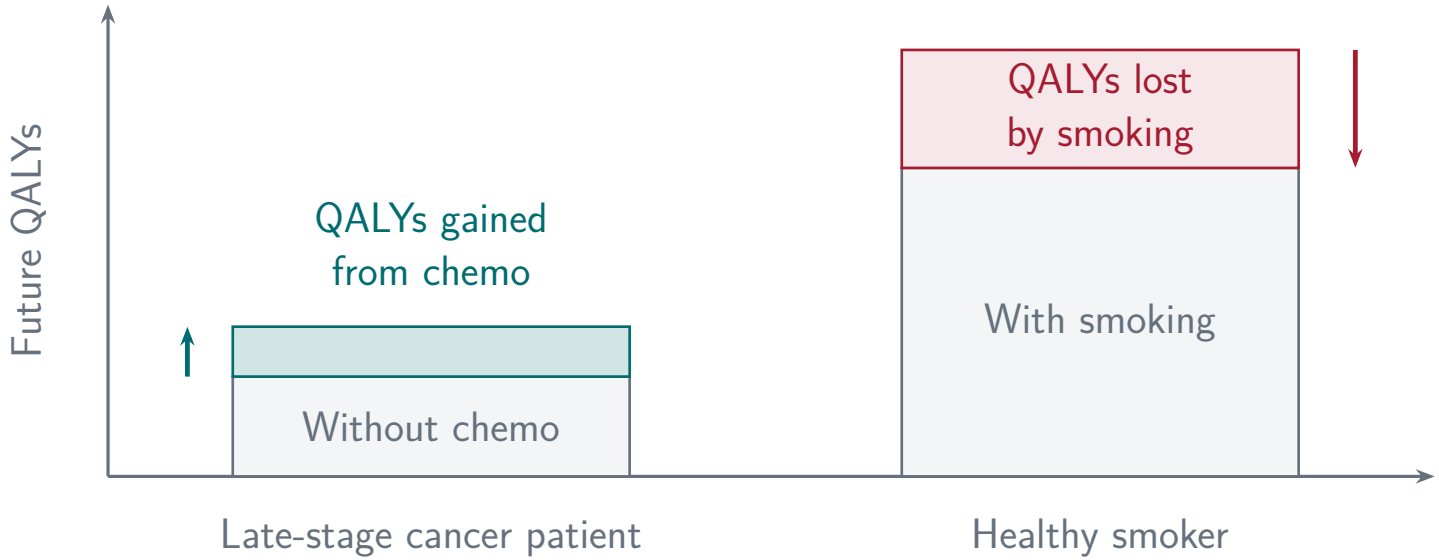
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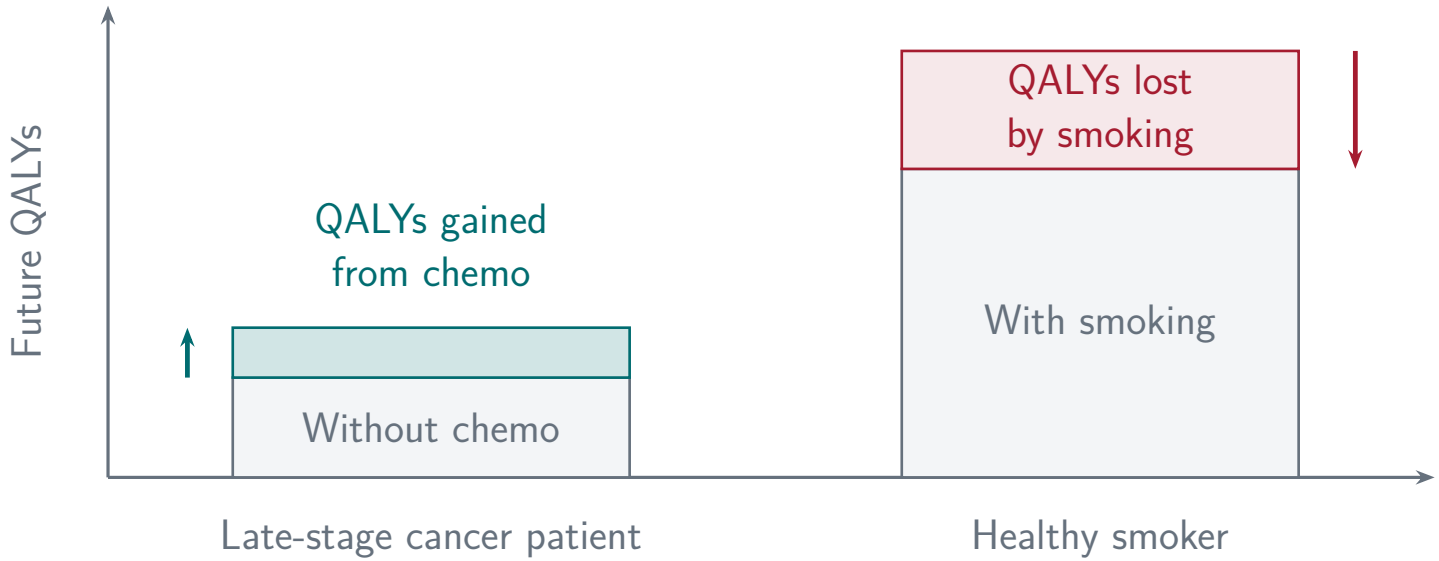
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- Maximizing QALY gains can therefore disadvantage people with lasting disabilities or chronic illness.
- Individuals with financial means are expected to both live longer and to do so in better health, e.g., due to better access to healthcare.
- QALY metric doesn't accurately capture certain widespread patterns of human preferences.

Preferences and QALYs

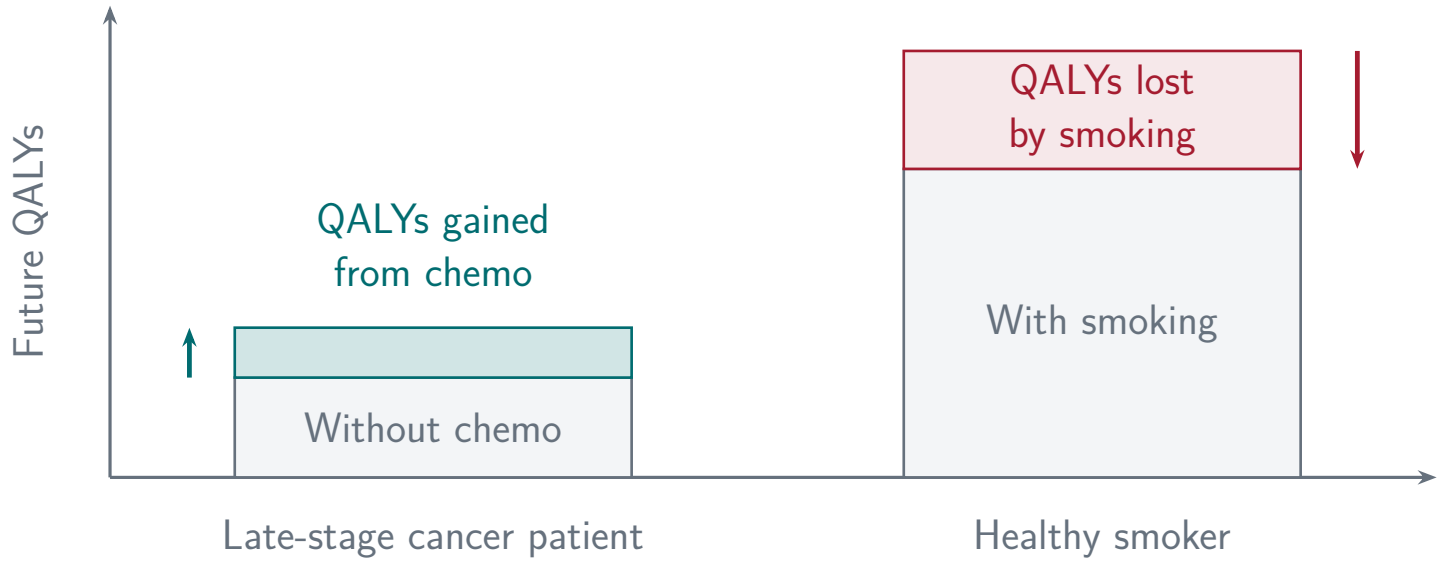


Preferences and QALYs



Interpretation: cancer patient values a small number of gained QALYs greater than the otherwise healthy smoker values a fairly large number of lost QALYs.

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Generally, people tend to value small gains given to the worst off individuals as more valuable than large gains given to relatively better off individuals.

Class poll-1

If we had to choose one of the two following patients for a kidney donation, who should we choose?

- Patient A: Has lived 50 years in near-perfect health, and would gain 20 more years in near-perfect health with a kidney donation.
- Patient B: Has lived 20 years in near-perfect health, and would gain 19 more years in near-perfect health with a kidney donation.

Class poll-2

What about for the following two patients?

- Patient C: Has lived 20 years, most of which with childhood leukemia, which is now cured. Would gain 10 years in near-perfect health with a kidney donation.
- Patient D: Has lived 20 years in near-perfect health, and would gain 10 years in near-perfect health with a kidney donation.

Prioritarianism

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Past disadvantage may give us a reason to prioritize B over A, and C over D.

Prioritarianism: Rawls' Maximin

For each patient v, use two **expected lifetime welfare** levels:

$$w_0(v): \text{ without a kidney donation,}$$
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Maximin: make the worst-off patient as well off as possible. Try to achieve

$$\max_M \min\Big(\{w_0(v) : v \text{ unmatched}\} \cup \{w_1(v) : v \text{ matched}\}\Big).$$

Break ties using the second-worst-off patient, then the third, and so on.

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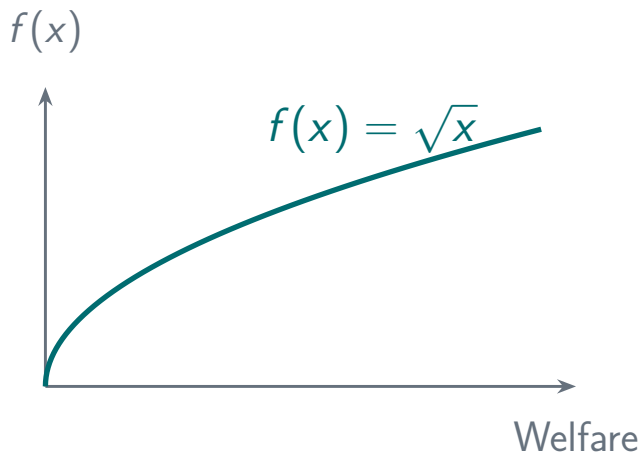
Concern: a tiny gain for the worst-off patient takes priority over very large gains for others.

Prioritarianism: Valuing Welfare

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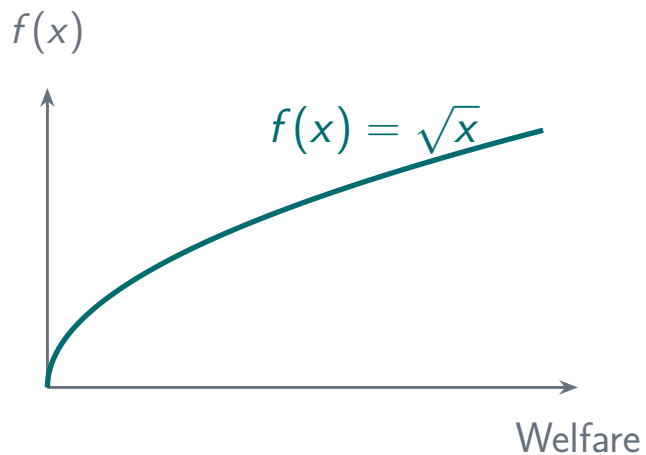


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For patients B and A in the example:

$$\underbrace{\sqrt{39} - \sqrt{20}}_{\text{B: gain of 19}} > \underbrace{\sqrt{70} - \sqrt{50}}_{\text{A: gain of 20}}.$$

Prioritarianism and Weighted Matching

Choose a matching M that maximizes

$$\sum_{\substack{v \text{ matched} \\ \text{patient}}} \big(f(w_1(v)) - f(w_0(v))\big).$$

- This is a 'robust version' of Rawls' maximin - to see it, add $\underbrace{\sum_{v \text{ patient}} f(w_0(v))}_{\text{independent of } M}$ and obtain the weight of a matching M:

$$\sum_{\substack{v \text{ unmatched} \\ \text{patient}}} f(w_0(v)) \; + \; \sum_{\substack{v \text{ matched} \\ \text{patient}}} f(w_1(v)).$$

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- Use **Maximum Vertex-Weighted Matching** with patient weights

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Choosing the welfare measure and the function f remains an ethical judgment.

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Are we solving the right problem?